Exercise sheet 6

Linear Optimization (311.160), WS 2026/27

Raphael Bleier 16 November 2026

Exercise 6.1 (On the simplex method)

Let B be a basis and $x_B = A_B^{-1}b - A_B^{-1}A_N x_N = \bar{b} - \bar{A}x_N$ the corresponding system of equations in the simplex dictionary obtained by splitting $Ax = b$, where $\bar{A}$ has entries $\bar{a}_{iq}$. Suppose that we choose x_{q_0} as entering variable and that the minimum ratio test gives us x_{i_0} as leaving variable (note that this means $\bar{a}_{i_0 q_0} > 0$). We denote by $\tilde{B}$ the set of indices given by changing q_0 and i_0 in B and solve the system $x_B = \bar{b} - \bar{A}x_N$ for $x_{\tilde{B}}$ to get a new dictionary. Show that $\tilde{B}$ is a basis.

Setup

The entering and leaving indices swap places:

$$\tilde{B} = (B \setminus \{i_0\}) \cup \{q_0\} \qquad \text{definition of } \tilde{B}$$
$$\tilde{N} = (N \setminus \{q_0\}) \cup \{i_0\} \qquad \text{the remaining variables}$$

To say that $\tilde{B}$ is a basis means that $A_{\tilde{B}}$ is invertible.

The new dictionary is equivalent to $Ax = b$

The old dictionary satisfies

$$x_B = A_B^{-1}b - A_B^{-1}A_N x_N \qquad \text{given form of the dictionary}$$
$$A_B x_B = b - A_N x_N \qquad \text{multiply by } A_B$$
$$A_B x_B + A_N x_N = b \qquad \text{move } A_N x_N \text{ to the left}$$

The last line is $Ax = b$, because B and N partition the column indices of A. So the old dictionary and $Ax = b$ have the same solution set. The pivot constructs the new dictionary with two operations on this system. First, the equation of row i_0,

$$x_{i_0} = \bar{b}_{i_0} - \sum_{q \in N} \bar{a}_{i_0 q}\, x_q \qquad \text{row } i_0 \text{ of the old dictionary,}$$

is solved for x_{q_0}:

$$x_{q_0} = \frac{\bar{b}_{i_0}}{\bar{a}_{i_0 q_0}} - \frac{1}{\bar{a}_{i_0 q_0}}\, x_{i_0} - \sum_{q \in N,\, q \neq q_0} \frac{\bar{a}_{i_0 q}}{\bar{a}_{i_0 q_0}}\, x_q \qquad \text{divide by } \bar{a}_{i_0 q_0} > 0, \text{ isolate } x_{q_0}$$

This is possible because $\bar{a}_{i_0 q_0} > 0$, in particular $\bar{a}_{i_0 q_0} \neq 0$. Second, this expression for x_{q_0} is substituted into the remaining rows and into the objective row. Solving one equation for one variable and substituting the result into the other equations are rearrangements that change the form of the system but not its solution set. Therefore the new dictionary $x_{\tilde{B}} = \tilde{b} - \tilde{A}x_{\tilde{N}}$ has the same solution set as the old one, and hence the same solution set as $Ax = b$.

Every pair from the new dictionary satisfies $Ax = b$

Take any pair $x_{\tilde{B}}, x_{\tilde{N}}$ obtained from the system of equations in the new dictionary, that is, $x_{\tilde{B}} = \tilde{b} - \tilde{A}x_{\tilde{N}}$. This pair solves the new dictionary's system, and the new dictionary has the same solution set as $Ax = b$. Hence

$$A_{\tilde{B}} x_{\tilde{B}} + A_{\tilde{N}} x_{\tilde{N}} = b \qquad \text{the pair solves } Ax = b$$

because $\tilde{B}$ and $\tilde{N}$ also partition the column indices of A.

For each $x_{\tilde{N}}$ the completion $x_{\tilde{B}}$ is unique

Fix a value of $x_{\tilde{N}}$ and suppose both $x_{\tilde{B}}$ and $x'_{\tilde{B}}$ satisfy

$$\begin{aligned} A_{\tilde{B}}x_{\tilde{B}} &= b - A_{\tilde{N}}x_{\tilde{N}} && \text{assumption for } x_{\tilde{B}} \\ A_{\tilde{B}}x'_{\tilde{B}} &= b - A_{\tilde{N}}x_{\tilde{N}} && \text{assumption for } x'_{\tilde{B}} \end{aligned}$$

Then both pairs $(x_{\tilde{B}}, x_{\tilde{N}})$ and $(x'_{\tilde{B}}, x_{\tilde{N}})$ solve $Ax = b$, so both solve the new dictionary's system. But in the new dictionary the basic part is determined by the nonbasic part alone:

$$x_{\tilde{B}} = \tilde{b} - \tilde{A}x_{\tilde{N}} \qquad \text{form of the new dictionary}$$

and the right-hand side is the same number for both pairs. Hence $x_{\tilde{B}} = x'_{\tilde{B}}$. So for each $x_{\tilde{N}}$ there is exactly one solution $x_{\tilde{B}}$ of $A_{\tilde{B}}x_{\tilde{B}} = b - A_{\tilde{N}}x_{\tilde{N}}$.

$A_{\tilde{B}}$ is invertible

We use the equivalence from the hint: $A_{\tilde{B}}$ is invertible if and only if for each $x_{\tilde{N}}$ there is a unique solution $x_{\tilde{B}}$ of $A_{\tilde{B}}x_{\tilde{B}} = b - A_{\tilde{N}}x_{\tilde{N}}$. First direction. Assume $A_{\tilde{B}}$ is invertible. For each $x_{\tilde{N}}$, the vector

$$x_{\tilde{B}} = A_{\tilde{B}}^{-1}\big(b - A_{\tilde{N}}x_{\tilde{N}}\big) \qquad \text{apply } A_{\tilde{B}}^{-1}$$

solves the system. If $x'_{\tilde{B}}$ also solves it, then

$$\begin{aligned} A_{\tilde{B}}x'_{\tilde{B}} &= b - A_{\tilde{N}}x_{\tilde{N}} && \text{assumption} \\ A_{\tilde{B}}^{-1}A_{\tilde{B}}x'_{\tilde{B}} &= A_{\tilde{B}}^{-1}\big(b - A_{\tilde{N}}x_{\tilde{N}}\big) && \text{multiply by } A_{\tilde{B}}^{-1} \text{ from the left} \\ x'_{\tilde{B}} &= A_{\tilde{B}}^{-1}\big(b - A_{\tilde{N}}x_{\tilde{N}}\big) && A_{\tilde{B}}^{-1}A_{\tilde{B}} = I \end{aligned}$$

so the solution is unique. Second direction. Assume that for each $x_{\tilde{N}}$ the system has a unique solution; existence is guaranteed by the first property, for example $x_{\tilde{B}} = \tilde{b} - \tilde{A}x_{\tilde{N}}$ for the pair with the same $x_{\tilde{N}}$. Suppose $A_{\tilde{B}}$ were not invertible. Then there is a vector $y \neq 0$ with

$$A_{\tilde{B}}y = 0 \qquad \text{nonzero null space of a singular matrix}$$

Take $x_{\tilde{N}} = 0$ and let $x_{\tilde{B}}$ be its unique solution. Then

$$\begin{aligned} A_{\tilde{B}}(x_{\tilde{B}} + y) &= A_{\tilde{B}}x_{\tilde{B}} + A_{\tilde{B}}y && \text{distribute } A_{\tilde{B}} \\ &= b - A_{\tilde{N}} \cdot 0 + 0 && \text{solution property and } A_{\tilde{B}}y = 0 \\ &= b - A_{\tilde{N}} \cdot 0 && \text{collect} \end{aligned}$$

so $x_{\tilde{B}} + y$ is a second, different solution for the same $x_{\tilde{N}} = 0$. This contradicts uniqueness. Hence $A_{\tilde{B}}$ is invertible, and $\tilde{B}$ is a basis.

$\boxed{\tilde{B} \text{ is a basis. } \square}$

Exercise 6.2 (Two-phase simplex)

Consider the problem: maximize $-x_1 - x_2$ subject to $2x_1 + x_2 \geq 3$, $x_1 + 2x_2 \geq 3$, $x \geq 0$. Set up the auxiliary problem and solve it. Is the original LP feasible? If so, what is the starting solution for the simplex algorithm?

Standard form and auxiliary problem

Introduce surplus variables $x_3, x_4 \geq 0$:

$$\begin{aligned} 2x_1 + x_2 - x_3 &= 3 && \text{first constraint, surplus } x_3 \\ x_1 + 2x_2 - x_4 &= 3 && \text{second constraint, surplus } x_4 \end{aligned}$$

Splitting $Ax = b$ with the slack basis $B = \{3, 4\}$ gives

$$\begin{aligned} x_3 &= -3 + 2x_1 + x_2 && \text{solve the first equation for } x_3 \\ x_4 &= -3 + x_1 + 2x_2 && \text{solve the second equation for } x_4 \end{aligned}$$

At $x_1 = x_2 = 0$:

$$\begin{aligned} x_3 &= -3 && \text{set } x_N = 0 \\ -3 &< 0 && \text{the value of } x_3 \\ -3 &< 0 && \text{the value of } x_4 \end{aligned}$$

The basic solution is infeasible, so the simplex method cannot start here. Add artificial variables $x_5, x_6 \geq 0$ and solve the auxiliary problem

$$\begin{aligned} & & \min\ w = x_5 + x_6 & && \text{drive both artificials to zero} \\ \text{subject to} & & 2x_1 + x_2 - x_3 + x_5 &= 3 && \text{artificial } x_5 \text{ in the first row} \\ & & x_1 + 2x_2 - x_4 + x_6 &= 3 && \text{artificial } x_6 \text{ in the second row} \\ & & x &\geq 0 && \text{all variables nonnegative} \end{aligned}$$

With the artificial basis $B = \{5, 6\}$ the dictionary is

$$\begin{aligned} x_5 &= 3 - 2x_1 - x_2 + x_3 && \text{solve row 1 for } x_5 \\ x_6 &= 3 - x_1 - 2x_2 + x_4 && \text{solve row 2 for } x_6 \end{aligned}$$

At $x_N = 0$ both artificials equal $3 \geq 0$, so this basis is feasible. In terms of the nonbasic variables,

$$\begin{aligned} z &= -w && \text{maximize } -w \text{ instead of minimizing } w \\ &= -x_5 - x_6 && \text{definition of } w \\ &= -(3 - 2x_1 - x_2 + x_3) - (3 - x_1 - 2x_2 + x_4) && \text{substitute both rows} \\ &= -6 + 2x_1 + x_2 - x_3 + 3x_1 + 2x_2 - x_4 && \text{distribute the minus signs} \\ &= -6 + 3x_1 + 3x_2 - x_3 - x_4 && \text{collect terms} \end{aligned}$$

Phase 1, first pivot

The reduced costs of (x_1, x_2, x_3, x_4) are $(3, 3, -1, -4)$. The largest positive value 3 occurs twice; the tie goes to the earlier position in $N = (1, 2, 3, 4)$, so x_1 enters.

$$\begin{aligned} 3 &= 3 && \text{tie between } x_1 \text{ and } x_2 \\ 1 &< 2 && \text{the earlier index wins} \end{aligned}$$

The column of x_1 in the dictionary is $(2, 1)^T$. Both entries are positive, so both rows limit the step:

$$\begin{aligned} \frac{3}{2} &= \tfrac{3}{2} && \text{ratio in the } x_5 \text{ row} \\ \frac{3}{1} &= 3 && \text{ratio in the } x_6 \text{ row} \\ \tfrac{3}{2} &< 3 && \text{compare the ratios} \end{aligned}$$

x_5 leaves with step length $\frac{3}{2}$. Solve its row for x_1:

$$\begin{aligned} x_5 &= 3 - 2x_1 - x_2 + x_3 && \text{old row} \\ 2x_1 &= 3 - x_5 - x_2 + x_3 && \text{move } 2x_1 \text{ left, } x_5 \text{ right} \\ x_1 &= \tfrac{3}{2} - \tfrac{1}{2}x_2 + \tfrac{1}{2}x_3 - \tfrac{1}{2}x_5 && \text{divide by 2} \end{aligned}$$

Substitute into the remaining rows:

$$\begin{aligned} x_6 &= 3 - x_1 - 2x_2 + x_4 && \text{old row} \\ &= 3 - \left(\tfrac{3}{2} - \tfrac{1}{2}x_2 + \tfrac{1}{2}x_3 - \tfrac{1}{2}x_5\right) - 2x_2 + x_4 && \text{substitute } x_1 \\ &= 3 - \tfrac{3}{2} + \tfrac{1}{2}x_2 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_5 - 2x_2 + x_4 && \text{distribute the minus} \\ &= \tfrac{3}{2} - \tfrac{3}{2}x_2 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_5 + x_4 && \text{collect terms} \\ z &= -6 + 3x_1 + 3x_2 - x_3 - x_4 && \text{old objective} \\ &= -6 + 3\left(\tfrac{3}{2} - \tfrac{1}{2}x_2 + \tfrac{1}{2}x_3 - \tfrac{1}{2}x_5\right) + 3x_2 - x_3 - x_4 && \text{substitute } x_1 \\ &= -6 + \tfrac{9}{2} - \tfrac{3}{2}x_2 + \tfrac{3}{2}x_3 - \tfrac{3}{2}x_5 + 3x_2 - x_3 - x_4 && \text{distribute the 3} \\ &= -\tfrac{3}{2} + \tfrac{3}{2}x_2 + \tfrac{1}{2}x_3 - \tfrac{3}{2}x_5 - x_4 && \text{collect terms} \end{aligned}$$

The new basic solution is $(x_1, x_2, x_3, x_4, x_5, x_6) = (\frac{3}{2}, 0, 0, 0, 0, \frac{3}{2})$ with $z = -\frac{3}{2}$.

Phase 1, second pivot

The reduced costs of (x_2, x_3, x_4, x_5) in the new objective are $(\frac{3}{2}, \frac{1}{2}, -\frac{3}{2}, -1)$. Only positive ones qualify:

$$\begin{aligned} \tfrac{1}{2} &> 0 && x_3 \text{ qualifies} \\ \tfrac{3}{2} &> \tfrac{1}{2} && x_2 \text{ has the larger value} \end{aligned}$$

x_2 enters. Its column is $(\frac{1}{2}, \frac{3}{2})^T$:

$$\begin{aligned} \frac{3/2}{1/2} &= 3 && \text{ratio in the } x_1 \text{ row} \\ \frac{3/2}{3/2} &= 1 && \text{ratio in the } x_6 \text{ row} \\ 1 &< 3 && \text{compare the ratios} \end{aligned}$$

x_6 leaves with step length 1. Solve its row for x_2:

$$\begin{aligned} x_6 &= \tfrac{3}{2} - \tfrac{3}{2}x_2 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_5 + x_4 && \text{old row} \\ \tfrac{3}{2}x_2 &= \tfrac{3}{2} - x_6 - \tfrac{1}{2}x_3 + \tfrac{1}{2}x_5 + x_4 && \text{move } \tfrac{3}{2}x_2 \text{ left, } x_6 \text{ right} \\ x_2 &= 1 - \tfrac{1}{3}x_3 + \tfrac{2}{3}x_4 + \tfrac{1}{3}x_5 - \tfrac{1}{3}x_6 && \text{divide by } \tfrac{3}{2} \end{aligned}$$

Substitute into the remaining rows:

$$\begin{aligned} x_1 &= \tfrac{3}{2} - \tfrac{1}{2}x_2 + \tfrac{1}{2}x_3 - \tfrac{1}{2}x_5 && \text{old row} \\ &= \tfrac{3}{2} - \tfrac{1}{2}\left(1 - \tfrac{1}{3}x_3 + \tfrac{2}{3}x_4 + \tfrac{1}{3}x_5 - \tfrac{1}{3}x_6\right) + \tfrac{1}{2}x_3 - \tfrac{1}{2}x_5 && \text{substitute } x_2 \\ &= \tfrac{3}{2} - \tfrac{1}{2} + \tfrac{1}{6}x_3 - \tfrac{1}{3}x_4 - \tfrac{1}{6}x_5 + \tfrac{1}{6}x_6 + \tfrac{1}{2}x_3 - \tfrac{1}{2}x_5 && \text{distribute } -\tfrac{1}{2} \\ &= 1 + \tfrac{2}{3}x_3 - \tfrac{1}{3}x_4 - \tfrac{2}{3}x_5 + \tfrac{1}{3}x_6 && \text{collect terms} \\ z &= -\tfrac{3}{2} + \tfrac{3}{2}x_2 + \tfrac{1}{2}x_3 - \tfrac{3}{2}x_5 - x_4 && \text{old objective} \\ &= -\tfrac{3}{2} + \tfrac{3}{2}\left(1 - \tfrac{1}{3}x_3 + \tfrac{2}{3}x_4 + \tfrac{1}{3}x_5 - \tfrac{1}{3}x_6\right) + \tfrac{1}{2}x_3 - \tfrac{3}{2}x_5 - x_4 && \text{substitute } x_2 \\ &= -\tfrac{3}{2} + \tfrac{3}{2} - \tfrac{1}{2}x_3 + x_4 + \tfrac{1}{2}x_5 - \tfrac{1}{2}x_6 + \tfrac{1}{2}x_3 - \tfrac{3}{2}x_5 - x_4 && \text{distribute } \tfrac{3}{2} \\ &= -x_5 - x_6 && \text{collect terms} \end{aligned}$$

Optimality of phase 1 and the starting solution

The final dictionary has $B = \{1, 2\}$, $N = \{3, 4, 5, 6\}$ and reduced costs $(0, 0, -1, -1)$:

$$\begin{aligned} 0 &\leq 0 && \text{reduced cost of } x_3 \\ 0 &\leq 0 && \text{reduced cost of } x_4 \\ -1 &\leq 0 && \text{reduced cost of } x_5 \\ -1 &\leq 0 && \text{reduced cost of } x_6 \end{aligned}$$

So $z = -w$ is maximal. At $x_N = 0$:

$$\begin{aligned} z &= -x_5 - x_6 && \text{final objective row} \\ &= 0 - 0 && \text{set } x_N = 0 \\ &= 0 && \text{compute the difference} \end{aligned}$$

The maximum of $-w$ is 0, hence $\min w = 0$ with $x_5 = x_6 = 0$. The original LP is feasible.

$$\boxed{\text{the original LP is feasible}}$$

Dropping the artificials x_5, x_6 from the dictionary leaves the starting dictionary for the simplex algorithm on the original problem:

$$\begin{aligned} x_1 &= 1 + \tfrac{2}{3}x_3 - \tfrac{1}{3}x_4 && \text{set } x_5 = x_6 = 0 \text{ in the } x_1 \text{ row} \\ x_2 &= 1 - \tfrac{1}{3}x_3 + \tfrac{2}{3}x_4 && \text{set } x_5 = x_6 = 0 \text{ in the } x_2 \text{ row} \\ z &= -x_1 - x_2 && \text{original objective} \\ &= -2 - \tfrac{1}{3}x_3 - \tfrac{1}{3}x_4 && \text{substitute both rows} \end{aligned}$$

The starting basic feasible solution is $x = (1, 1, 0, 0)^T$ with basis $B = \{1, 2\}$.

$$\boxed{x = (1, 1, 0, 0)^T, \quad B = \{1, 2\}}$$

Exercise 6.3 (Programming)

We continue with Exercise 5.3 and Example 2.2 of the lecture notes. Rename `initial_simplex_dictionary` *to* `create_simplex_dictionary`*. Use* `create_simplex_dictionary`*,* `choose_entering_variable_index` *and* `choose_leaving_variable_index` *to write an algorithm that solves the problem of Example 2.2 numerically. Terminate when all reduced costs in a dictionary are nonpositive. Test your algorithm with the data of Example 2.2. Submit* `LASTNAME_sheet_NUMBER.py` *and* `LASTNAME_sheet_NUMBER.pdf`*.*

Why one dictionary function is enough

Exercise 6.1 shows that the new dictionary can be obtained in two equivalent ways: by solving the equations of the old dictionary for the new basis $\tilde{B}$, or by splitting $Ax = b$ directly as $x_{\tilde{B}} = A_{\tilde{B}}^{-1}b - A_{\tilde{B}}^{-1}A_{\tilde{N}}x_{\tilde{N}}$. The second way is exactly what `initial_simplex_dictionary` from sheet 4 computes. So the pivot needs no new function, only a new basis:

$$\tilde{B} = (B \setminus \{i_0\}) \cup \{q_0\} \qquad \text{replace leaving index } i_0 \text{ by entering index } q_0$$

and a fresh call to the renamed function.

The algorithm

The function `solve_simplex` in `Bleier_sheet_6.py` follows the pseudocode of the lecture notes:

1. Create the dictionary for the starting basis B and reject a basis with $\bar{b} < 0$ components, since the loop has no phase 1.
2. Ask `choose_entering_variable_index`. If it returns `None`, all reduced costs are nonpositive and the current basic solution is optimal.
3. Otherwise ask `choose_leaving_variable_index`. If it returns `None`, the entering column has no positive entry and the LP is unbounded.
4. Replace the leaving index by the entering index at its position in B and go back to step 1.

A counter bounds the number of iterations, so a cycling basis cannot loop forever.

Test on Example 2.2

Example 2.2 is $\max -x_1 + 2x_2 + 6x_3$ with

$$
\begin{aligned}
2x_1 + 3x_2 + x_4 &= 10 && \text{first constraint} \\
x_1 + 2x_3 + x_5 &= 8 && \text{second constraint} \\
x_1 + 2x_2 + x_3 + x_6 &= 6 && \text{third constraint}
\end{aligned}
$$

The run starts from the slack basis $B = (4, 5, 6)$:

$$
\begin{aligned}
\bar{c}_N &= (-1, 2, 6)^T && \text{initial reduced costs} \\
N &= (1, 2, 3) && \text{nonbasis of the first dictionary}
\end{aligned}
$$

x_3 enters, x_5 leaves ratio test: $\frac{8}{2} = 4 < \frac{6}{1} = 6$

The second dictionary has $B = (4, 3, 6)$, $\bar{b} = (10, 4, 2)^T$, $z = 24$ and

$$
\begin{aligned}
\bar{c}_N &= (-4, 2, -3)^T && \text{second reduced costs} \\
N &= (1, 2, 5) && \text{nonbasis of the second dictionary}
\end{aligned}
$$

x_2 enters, x_6 leaves ratio test: $\frac{2}{2} = 1 < \frac{10}{3}$

The third dictionary has $B = (4, 3, 2)$, $\bar{b} = (7, 4, 1)^T$, $z = 26$ and

$$
\begin{aligned}
\bar{c}_N &= (-\tfrac{9}{2}, -\tfrac{5}{2}, -1)^T && \text{third reduced costs} \\
-\tfrac{9}{2} &\le 0 && \text{reduced cost of } x_1 \\
-\tfrac{5}{2} &\le 0 && \text{reduced cost of } x_5 \\
-1 &\le 0 && \text{reduced cost of } x_6 \\
N &= (1, 5, 6) && \text{nonbasis of the third dictionary}
\end{aligned}
$$

The loop stops after two pivots. The solution is

$$
\begin{aligned}
x &= (0, 1, 4, 7, 0, 0)^T && \text{basic values from } \bar{b} \\
z &= 26 && \text{objective value at this basis}
\end{aligned}
$$

$$
\boxed{x = (0, 1, 4, 7, 0, 0)^T, \quad z = 26}
$$